

INDIVIDUAL PROJECT (LSDS)

WEEK 4 REPORT

(Hive Implementation & Analytical Queries)

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1) Objective of Week 4:

The objective of Week 4 was to perform large-scale data analysis on the automobile dataset stored in HDFS using Apache Hive. Hive was used to create structured tables and execute SQL-based analytical queries on big data.

2) Hive Database Creation:

A new database was created for the project.

```
CREATE DATABASE car_analysis;
```

The database was activated using:

```
USE car_analysis;
```

3) Hive Table Creation:

An external table was created to read the automobile dataset directly from HDFS.

Table:

```

hive> USE car_analysis;
OK
Time taken: 0.976 seconds
hive> CREATE EXTERNAL TABLE car_prices (
    >     car_id INT,
    >     brand STRING,
    >     year INT,
    >     engine_size DOUBLE,
    >     fuel_type STRING,
    >     transmission STRING,
    >     mileage INT,
    >     condition STRING,
    >     price DOUBLE,
    >     model STRING
    > )
    > ROW FORMAT DELIMITED
    > FIELDS TERMINATED BY ','
    > STORED AS TEXTFILE
    > LOCATION '/user/maria_dev/car_project/';
OK
Time taken: 183.635 seconds

```

4) Verification of Table:

Table structure was verified using:

DESCRIBE car_prices;

Sample data was checked using:

SELECT * FROM car_prices LIMIT 5;

This confirmed successful integration of Hive with HDFS data.

```

hive> DESCRIBE car_prices;
OK
car_id          int
brand           string
year            int
engine_size     double
fuel_type       string
transmission    string
mileage         int
condition       string
price           double
model           string
Time taken: 0.656 seconds, Fetched: 10 row(s)

```

```
hive> SELECT * FROM car_prices LIMIT 5;
OK
NULL    Brand    NULL    NULL    Fuel Type    Transmission    NULL    Condition    NULL
odel
1    Tesla    2016    2.3    Petrol Manual    114832    New    26613.92    Model X
2    BMW    2018    4.4    Electric    Manual    143190    Used    14679.61    5 Ser
ies
3    Audi    2013    4.5    Electric    Manual    181601    New    44402.61    A4
4    Tesla    2011    4.1    Diesel Automatic    68682    New    86374.33    Model
Y
Time taken: 0.493 seconds, Fetched: 5 row(s)
```

5) Analytical Queries & Insights:

Query 1: Average Price Per Brand:

```
hive> SELECT brand, AVG(price) AS avg_price
> FROM car_prices
> GROUP BY brand
> ORDER BY avg_price DESC;
Query ID = maria_dev_20260302065538_d85d64ef-ec10-4f94-a8db-9d10c6a4718c
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1772434087936_0001)
```

	VERTICES	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	SUCCEEDED	10	10	0	0	0	0	
Reducer 2	SUCCEEDED	1	1	0	0	0	0	
Reducer 3	SUCCEEDED	1	1	0	0	0	0	

```
VERTICES: 03/03  [=====>>] 100% ELAPSED TIME: 70.57 s
-----
OK
BMW    54157.114385475754
Tesla  53475.547471262485
Mercedes  53191.09008498343
Toyota  52078.728235292234
Honda  52050.283948863376
Audi    51953.42480978109
Ford    51593.25481267795
NULL    NULL
Brand    NULL
Time taken: 74.317 seconds, Fetched: 9 row(s)
```

Insight:

BMW has the highest average price, followed by Tesla and Mercedes. Luxury brands show higher pricing trends.

Query 2: Count of Cars by Fuel Type:

```
hive> SELECT fuel_type, COUNT(*) AS total_cars
> FROM car_prices
> GROUP BY fuel_type;
Query ID = maria_dev_20260302065703_2abbd6df-0362-4c4a-b10b-aa3ba285fef3
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1772434087936_0001)

-----
      VERTICES      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 .....  SUCCEEDED    10         10         0         0         0         0
Reducer 2 .....  SUCCEEDED     1          1         0         0         0         0
-----
VERTICES: 02/02  [=====>>] 100%  ELAPSED TIME: 37.62 s
-----

OK
NULL      7
Diesel    655000
Electric      614000
Fuel Type      1
Hybrid    601000
Petrol    630000
Time taken: 42.297 seconds, Fetched: 6 row(s)
```

Insight:

This analysis shows the distribution of vehicles based on fuel types such as petrol, diesel, hybrid, or electric.

Query 3: Average Price by Transmission:

```
hive> SELECT transmission, AVG(price) AS avg_price
> FROM car_prices
> GROUP BY transmission;
Query ID = maria_dev_20260302065804_501a61d4-7558-432b-8c0b-e1d18e878e08
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1772434087936_0001)

-----
      VERTICES      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 .....  SUCCEEDED    10         10         0         0         0         0
Reducer 2 .....  SUCCEEDED     1          1         0         0         0         0
-----
VERTICES: 02/02  [=====>>] 100%  ELAPSED TIME: 29.94 s
-----

OK
NULL      NULL
Automatic    52691.684639259984
Manual    52589.11944954217
Transmission      NULL
Time taken: 31.951 seconds, Fetched: 4 row(s)
```

Insight:

Automatic transmission vehicles generally have higher average pricing compared to manual transmission vehicles.

Query 4: Average Price by Manufacturing Year:

```
hive> SELECT year, AVG(price) AS avg_price
> FROM car_prices
> GROUP BY year
> ORDER BY year;
Query ID = maria_dev_20260302065843_dad1fa67-fb64-47d2-967d-fae7880a8cea
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1772434087936_0001)
```

	VERTICES	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1		SUCCEEDED	10	10	0	0	0	0
Reducer 2		SUCCEEDED	1	1	0	0	0	0
Reducer 3		SUCCEEDED	1	1	0	0	0	0

```
VERTICES: 03/03 [=====>>] 100% ELAPSED TIME: 27.85 s
```

```
OK
NULL NULL
2000 53094.163440860255
2001 48944.05770642213
2002 57197.47599999993
2003 52260.187372881635
2004 51517.97689999878
2005 57534.32359223286
2006 53437.59593023152
2007 56464.203545454686
2008 53104.78871559694
2009 52932.73577777839
2010 53650.24463158004
2011 51414.87118811891
2012 57092.138378377735
2013 52495.03473684052
2014 54202.5772477075
2015 51516.460909090674
2016 51977.464067796354
2017 49045.6891752574
2018 52640.94660377403
2019 49848.33902912594
2020 48281.73942623035
2021 48819.71508928673
2022 54656.13955357041
2023 51328.69271844691
Time taken: 32.679 seconds, Fetched: 25 row(s)
hive>
```

Insight:

Newer manufacturing years show higher average prices, demonstrating vehicle depreciation trends in the automobile market.

6) Technical Observations:

- Dataset Size Processed: ~152MB
- Records Processed: 2.5 Million
- Hive executed queries using distributed processing via YARN
- Aggregations performed efficiently on large-scale data

7) Conclusion:

Hive successfully enabled structured querying and large-scale analytical processing of automobile pricing data stored in HDFS.

The analysis provided meaningful business insights such as brand-wise pricing trends, fuel distribution, transmission impact, and year-based depreciation patterns.

This demonstrates how Hive simplifies big data analytics using SQL-like queries on distributed systems.